Felipe Arthur Eckert

Data & Software Developer | Python · Data Pipelines · Machine Learning · Full-stack Apps

Indaiatuba, SP, Brazil | felipeeckert10@gmail.com | +55 19 99858-4297

LinkedIn | GitHub

PROFESSIONAL SUMMARY

Electrical engineer from Poli-USP who builds data products end to end. At Hyundai Motor Brasil I designed and shipped an internal analysis app, from raw-file ingestion and validation to the statistical model and automated reports; it cut analysis time by about 80% and became a paper at SIMEA 2026. On my own I have shipped a full-stack web app with a REST API and PostgreSQL, and an ML model benchmarked against deep learning.

SKILLS

Languages & Frameworks: Python (advanced), pandas, NumPy, SQL (PostgreSQL, SQLite), TypeScript, React, Next.js, Streamlit, C/C++

Data & ML: ETL pipelines, data cleaning and validation, statistics, scikit-learn, XGBoost, PyTorch, Optuna, SHAP

Engineering: Git/GitHub, CI/CD with GitHub Actions, automated testing (pytest, Vitest, Playwright), REST APIs, Vercel

AI-assisted Dev: Claude Code agentic workflows, sub-agents, custom skills and hooks, Microsoft 365 Copilot

Domain: Test, sensor and time-series data, embedded data acquisition, automotive R&D; quick to learn new domains

WORK EXPERIENCE

Hyundai Motor Brasil *Jan 2025 – Present*

Junior Research & Development Analyst – Energy Efficiency & Data Analytics

- Designed and shipped an internal coastdown analysis app end to end (Python, Streamlit, SQLite): multi-source ingestion, weather-data sync, automated validation, statistical model and standardized reports. Cut analysis time by about 80%; published at SIMEA 2026.
- Maintained it through ~150 commits with 500+ automated tests and a performance audit, using Claude Code as a coding assistant.
- Developed internal analytics and visualization tools (Python, Plotly, Streamlit) adopted by the engineering team, replacing manual spreadsheet workflows.
- Won 1st place at the Hyundai Prompt-a-thon (Sep 2026) with R&D colleagues: a Microsoft 365 Copilot hub that consolidates multi-source information into executive reports.

Poli-USP BAJA SAE Team *Mar 2018 – Dec 2024*

Powertrain Director (2021 – 2024) · Powertrain Coordinator (2018 – 2021)

- Co-developed a custom STM32 telemetry data logger for vehicle setup decisions (topic of my undergraduate thesis).
- Led the team's first 4x4 gearbox from concept to competition, coordinating suppliers on cost and lead time.

SELECTED PROJECTS

- Paper trading simulator (live demo): full-stack Next.js and PostgreSQL app with auth, a REST API (API keys, rate limits, idempotent orders), a backtester and ~220 tests in CI; deployed on Vercel.
- Vehicle energy efficiency estimator (GitHub): ETL over Brazil's PBEV 2026 label data, then XGBoost vs. random forest and PyTorch networks under grouped nested CV with Optuna and monotonic constraints (5.8% error); Streamlit app.
- Finale (GitHub): one-knob VST3 mastering plugin in C++ (JUCE), built with a friend: real-time audio engine, loudness metering, ~180 tests and CI.

EDUCATION

Escola Politécnica da USP (Poli-USP) *2018 – 2025*

B.S. in Electrical Engineering — Emphasis: Automation & Control | Specialization: Automotive Engineering

LANGUAGES

Portuguese: Native · **English:** Advanced (C1), professional working proficiency · **Spanish:** Conversational (B1)

PUBLICATIONS & AWARDS

- Co-author, “Data-Driven Coastdown Analysis”, SIMEA 2026 (Blucher Engineering Proceedings).
- 1st Place Overall & 2nd Place (Logic) — Brazilian Knowledge Olympiad (OBS)